

# CHE 205, Fall 2019, Exam 3

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Blackboard submission due date: Thu, 12/12/2019, 11:59 PM

## Note:

- This is a take-home exam. You are allowed to use any resources available to you; however you are NOT allowed to talk about it with anyone.
- During the course of problem solving, if you derive any equation or perform any simplification, make sure to type your equations or simplifications in a text box (use insert tab, then text box from the top menu in Excel) in the spreadsheet.
- This exam is worth 35 points, and 25% of your total grade.

**Problem 1 (10 pts).** Consider the following nonlinear equation:

$$\left(a + \frac{b}{x}\right) = \frac{d}{(x - c)}, \quad (1)$$

where  $x \geq 1$ ,  $a = 10$ ,  $b = 4.238$ ,  $c = 0.038$ , and  $d = 42.917$ . Find  $x$  for which equation 1 is satisfied, using the following steps:

- (a) Plot the function whose root you are trying to find. (2 pts)
- (b) Based on your results in part a, make a good initial guess,  $x_0$ . (1 pt)
- (c) Write a VBA script to find the root of the function in part a, using Newton's method, and the initial guess,  $x_0$ , you made in part b. The VBA script should take  $a$ ,  $b$ ,  $c$ ,  $d$ ,  $x_0$ , and the tolerance,  $tol$ , as inputs through the spreadsheet. Then it should output
  - i. the solution,  $x$ ,
  - ii. the total number of iterations, and
  - iii. the error at each iteration, as shown below. (7 pts)

|    | A        | B           | C | D | E         | F     |
|----|----------|-------------|---|---|-----------|-------|
| 1  | 10       | =a          |   |   | iteration | error |
| 2  | 4.238    | =b          |   |   | 1         | ???   |
| 3  | 0.038    | =c          |   |   | 2         | ???   |
| 4  | 42.917   | =d          |   |   | 3         | ???   |
| 5  | 1.00E-09 | =tolerance  |   |   | 4         | ???   |
| 6  | ???      | =x0         |   |   | 5         | ???   |
| 7  | ???      | =x          |   |   | 6         | ???   |
| 8  | ???      | =iterations |   |   | 7         | ???   |
| 9  |          |             |   |   | 8         | ???   |
| 10 |          |             |   |   |           |       |

Figure 1: Sample input and output of VBA script in Problem 1.

**Problem 2 (10 pts).** Using the Crank-Nicholson method, write a VBA program to solve the following diffusion equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - B \frac{\partial u}{\partial x}, \quad (2)$$

subject to the following initial condition:

$$u(x, t = 0) = \sin(\pi x), \quad (3)$$

and boundary conditions:

$$u(x, t) = 0 \text{ at } x = 0 \quad (4)$$

$$\frac{\partial u(x, t)}{\partial x} = 1 \text{ at } x = 1, \quad (5)$$

Set  $B = 1$ , and use  $t_{max}=1$ ,  $\Delta x=0.05$ , and  $\Delta t=0.001$  to plot  $u(x, t)$  against  $0 \leq x \leq 1$  for four different values of  $t=0$  (blue), 0.01 (green), 0.1 (red), and 1 (black).

**Problem 3 (10 pts).** A model for a batch reactor is given by:

$$\begin{aligned}\frac{dy}{dt} &= b_1 y - \frac{b_1}{b_2} y^2, & y(t=0) &= 0.03 \\ \frac{dz}{dt} &= b_3 y, & z(t=0) &= 0.0\end{aligned}$$

where

$y$  = dimensionless concentration of the first component

$z$  = dimensionless concentration of the second component

$t$  = dimensionless time,  $0 \leq t \leq 1$ .

Experiments have determined that  $b_1 = 13.1$ ,  $b_2 = 0.94$ , and  $b_3 = 1.71$ .

- (a) Solve this ordinary differential equation - initial value problem (ODE-IVP) using forward Euler in spreadsheet (this is not VBA programming) with  $\Delta t = 0.005$ . Plot  $y$  and  $z$  in red and blue, respectively, vs  $0 \leq t \leq 1$ . (3 pts)
- (b) Write a VBA program to get  $b_1$ ,  $b_2$ ,  $b_3$  and initial conditions from a spreadsheet as inputs, and solve this ODE-IVP using the 2nd-order RK. Plot  $y$  and  $z$  vs  $t$  as in part a. (7 pts)

**Problem 4 (5 pts).** A heterogeneous reaction is known to occur at a rate described by the following expression:

$$\log_{10}(r) = \log_{10}(k_1 P_A) - 2\log_{10}(1 + k_A P_A + k_B P_B), \quad (6)$$

where  $\log_{10}$  is the log base 10, and  $k_1 = 0.015$ . Using an optimization method and the following rate data, estimate the values of  $k_A$  and  $k_B$ .

| $P_A$ | $P_B$ | $r$      |
|-------|-------|----------|
| 1     | 0     | 3.40E-05 |
| 0.9   | 0.1   | 3.60E-05 |
| 0.8   | 0.2   | 3.70E-05 |
| 0.7   | 0.3   | 3.90E-05 |
| 0.6   | 0.4   | 4.00E-05 |

Table 1: Rate data for different values of  $P_A$  and  $P_B$ .